Linear Regression and OLS

Section 1: Simple Linear Regression and Residuals

1 In simple linear regression $\hat{y} = 3.5x + 12$, what is the residual $e_i = y_i - \hat{y}_i$ for an observation at $(x_i, y_i) = (4, 30)$?

a. $+4$

b. -4

c. $+2$

d. -2

2 What is the mathematical condition guaranteed by the OLS fit regarding the sum of all residuals $\sum_{i=1}^{n} e_i$?

a. $\sum e_i = 0$ exactly

b. $\sum e_i = 1$

c. $\sum e_i = n$

d. $\sum e_i = \text{SSE}$

3 Given $\bar{x} = 10$, $\bar{y} = 25$, and fitted slope $\hat{\beta}_1 = 2$, compute the y-intercept $\hat{\beta}_0$.

a. 5

b. 15

c. 20

d. 10

Section 2: Analytical OLS Formulas and Covariance

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4 For a dataset with $\text{Cov}(x, y) = 18$ and $\text{Var}(x) = 6$, compute the OLS slope $\hat{\beta}_1$.

a. 3.0

b. 0.33

c. 12

d. 108

5 If the Pearson correlation coefficient between x and y is $r = 0.8$, with $\sigma_x = 2$ and $\sigma_y = 5$, find the slope $\hat{\beta}_1$.

a. 2.0

b. 1.25

c. 0.32

d. 4.0

6 If all x values in a dataset are scaled by a factor of 10, how does the OLS regression slope change?

a. It is divided by 10

b. It is multiplied by 10

c. It is multiplied by 100

d. It remains unchanged

Section 3: Matrix Normal Equations and Projections

7 State the matrix Normal Equations for the OLS weight vector $\boldsymbol{\beta}$ given design matrix $\mathbf{X}$ and target $\mathbf{y}$.

a. $(\mathbf{X}^T\mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$

b. $\mathbf{X}\boldsymbol{\beta} = \mathbf{y}$

c. $(\mathbf{X}\mathbf{X}^T)\boldsymbol{\beta} = \mathbf{y}$

d. $\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$

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8 When is the Gram matrix ($\mathbf{X}^T\mathbf{X}$) strictly invertible?

a. When the columns of $\mathbf{X}$ are linearly independent (full column rank)

b. When $\mathbf{X}$ is a square matrix

c. When all elements in $\mathbf{X}$ are positive

d. When the target $\mathbf{y}$ has zero mean

9 What geometric role does the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ play?

a. It orthogonal-projects target $\mathbf{y}$ onto the column space of $\mathbf{X}$ ($\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$)

b. It inverts the covariance matrix

c. It computes feature cross-correlations

d. It normalizes features to zero mean

Section 4: Goodness of Fit and Regularization

10 In an evaluation, $\text{SSE} = 25$ and $\text{SST} = 100$. Compute the coefficient of determination R^2.

a. 0.75

b. 0.25

c. 0.50

d. 0.80

11 How does Ridge regression (L_2 penalty) resolve multicollinearity when $\mathbf{X}^T\mathbf{X}$ is ill-conditioned?

a. It adds $\lambda\mathbf{I}$ to $\mathbf{X}^T\mathbf{X}$, guaranteeing invertibility: $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}$

b. It drops half the features automatically

c. It sets all correlated weights to zero

d. It eliminates the bias term

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12 If $R^2 = 0.64$ in simple linear regression, what is the magnitude of the Pearson correlation coefficient $|r|$?

a. 0.80

b. 0.64

c. 0.32

d. 0.40

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